Date:13.01.2021

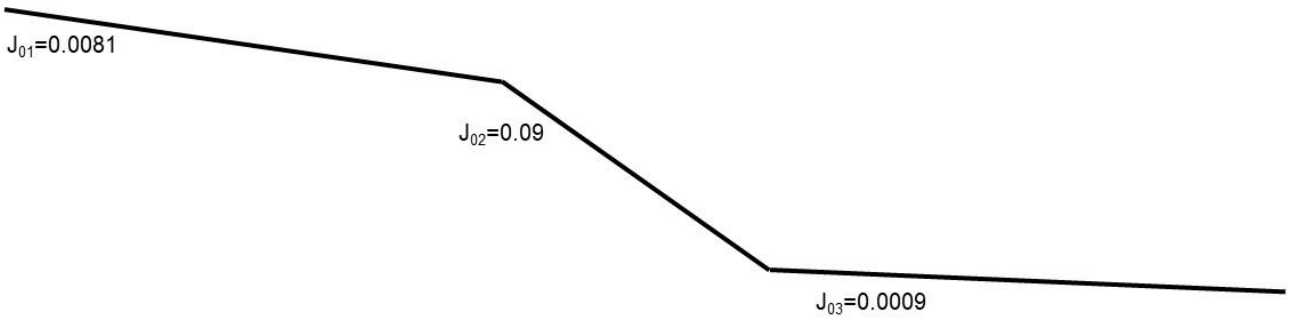
2020-2021 FALL SEMESTER HYDRAULICS COURSE FINAL EXAM

Name: **Surname:** **Student ID:**

Question 2 (30):

The width of the rectangular channel shown in the figure is B=(4.a) m, roughness n=0.014 and the discharge is Q=18 m^3/s. The slope of the channel changes as shown in the figure and are J_{01}=0.0081, J_{02}=0.09 ve J_{03}=0.0009.

B=(4.a) "a" is the last digit of your student ID.



a) Determine the critical water depth.
b) Determine the critical slope.
c) Determine the uniform flow depths in each part.
d) Identify all water surface profile typesthat can occur in each section. If there is a transition from the critical depth and a hydraulic jump occurs, calculate the conjugate depths and show on figure.
e) Draw the surface profile on the channel reach, write water surface profile types and all calculated values on the figure.